LESSON PLAN

AVIONICS AND ELECTRONIC WARFARE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	11
Total No. of Lesson Plans	25	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Earthing or grounding connections, cable terminations, crimped terminals and Its advantages.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Earthing or grounding connections, cable terminations, crimped terminals and Its advantages. Aluminum cable connections, Plugs and sockets. Different types of plugs and sockets potting, need for electrical Bonding. Static discharge wicks. Bonding and screening methods. Practical on crimping, bonding screening, earthing & grounding.
5. Specific Objectives	13.1.2.1 Knowledge of Static converting equipment, Bus bars, cable terminations, Switches.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Practical on crimping, bonding screening, earthing & grounding. Safety precautions: Verify power off, wear PPE, inspect tools. Instructor supervision: Monitor crimp quality, verify continuity, ensure correct torque.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of earthing in avionics?
(b) Why are crimped terminals preferred over soldered connections?
(c) What is the function of a static discharge wick?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Earthing and grounding connections • Aluminum cable connections • Cable terminations and crimped terminals • Advantages of crimped terminals	L & Ds	PPT, IB	15 min
(b)	• Plugs and sockets types • Potting requirements • Need for electrical bonding • Static discharge wicks	L & Ds	PPT, PDF	20 min
(c)	• Bonding and screening methods • Bus bar connections • Practical crimping exercise • Verification of bonding integrity	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between bonding and grounding in an aircraft system.
(b) Describe the procedure for verifying a crimped terminal connection.
(c) Why is potting necessary for certain avionics connectors?

12. Home Assignment and Follow Up	Prepare a brief note on the specific requirements for aluminum cable connections and how they differ from copper connections, referencing the next lesson on Switches.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Switches: - toggle switches, push switches, rocker button switches, rotary switches, micro switches, time s/w, mercury s/w, pressure s/w, thermal s/w, proximity s/w, rheostat and relays. Safeties to be taken while installing switches & relays. Practical switches & relays.	Refer to the master syllabus reference list	LP 12

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min